

2. Solution Framework

Logical Architecture

Nirvana operates on a four-tier architecture:

1. **Data Ingestion Layer:** Automated ETL scripts fetch live meteorological data (IMD), geographical data (Bhuvan/ISRO), and incident reports, storing them in a PostgreSQL Data Warehouse.
2. **Machine Learning Pipeline (Forecast Engine):** Predicts the 7-day demand using engineered features (`rainfall_7day` , `distance_from_coastline`). It outputs exact quantities needed for `affected_population` , `food` , `water` , `medical` , and `shelter` .
3. **Logistics Optimization Engine:** A mathematical solver takes the ML demand forecast and current NGO warehouse inventory as inputs. It formulates a Linear Programming problem to output the most efficient transport routes.
4. **Presentation Layer:** A responsive, interactive web application acting as the command center for NGO coordinators.

Process Flow (User Journey & Backend Execution)

- **Backend (Automated daily):**
 1. `data_ingestion/` scripts run via cron jobs, pushing data to PostgreSQL.
 2. `predict.py` executes an ensemble model (XGBoost/LightGBM/RF) to forecast demand metrics.
 3. `allocator.py` leverages Google OR-Tools to minimize transportation costs while maximizing fulfillment.
- **User Journey (NGO Coordinator):**
 1. **Log in:** Coordinator opens the Streamlit dashboard (`main.py`).
 2. **Assess:** Views the *Executive Overview* and *Prediction Dashboard* to identify high-risk districts (e.g., Category 4 cyclone alert).
 3. **Visualize:** Uses the *GIS Map* (Folium) to see color-coded risk zones.
 4. **Simulate:** Uses *Scenario Analysis* to “stress-test” inventory against worst-case weather events.
 5. **Execute:** Navigates to *Resource Allocation*, downloads the OR-Tools generated route manifest, and hands it to truck dispatchers.

Tech Stack

- **Backend & Data Processing:** Python 3.12, Pandas, Numpy, Scikit-Learn
- **Database:** PostgreSQL (with SQLAlchemy and psycopg2-binary)
- **Machine Learning:** XGBoost, LightGBM, Prophet (Time-series)
- **Logistics Optimization:** Google OR-Tools
- **Frontend UI & Visuals:** Streamlit, Plotly, Streamlit-Folium, Geopandas
- **Deployment:** Docker, Docker Compose

Financial Model: B2B SaaS / Freemium

- **Freemium Tier:** Free access to district-level risk overviews and basic weather alerts. Aimed at onboarding small, local NGOs and building user trust.
- **Enterprise License (B2B):** Paid subscriptions for large NGOs (e.g., Red Cross, Oxfam) and State Govts (SDMAs). Unlocks the Google OR-Tools optimization engine, private multi-warehouse inventory integration, and high-resolution localized forecasting.